Computational Modeling of Human Experience Formation through Practice: Evidence, Revision, Reuse, and Host-Agent Validation

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Abstract— Experience is not acquired merely by saving a piece of text. It emerges from goal-directed action, action outcomes, interpretation, counterexample testing, and subsequent reuse. Omitting any of these steps may turn an accidental success into an apparent rule. This paper models this human process as a computational cycle of practice, observation, interpretation, validation, experience, and renewed practice. Experience is represented by an object containing a claim, conditions, version, provenance, evidence, counterexamples, and lifecycle state. Candidate, stable, disputed, and invalidated knowledge are distinguished. Persistent version chains, provenance constraints, conflict records, explicit unknown outcomes, and host-context recovery allow applicability to be checked before the next action. Beta–Bernoulli updating describes changes in belief under limited evidence, information gain ranks informative verification actions, and Thompson sampling serves only as an optional exploration controller; the overall system is not characterized as Bayesian reinforcement learning. We first explain the experience-formation process, then present state transitions, update and recovery algorithms, and complexity, followed by mechanism experiments with 10 seeds, 12 rounds, and seven conditions, a 20-seed end-to-end gate experiment, and separate 10-seed Bayesian bandit controls. In the rule-reuse benchmark, B2 improves completion by 4.51 percentage points over an exact-index baseline and reduces source reads by 81.60%, while increasing local logical operations by 41.53% and providing no latency advantage. The end-to-end gate experiment places experience formation and task benefit in one ledger: with the candidate gate returned to the loop, completion remains 4.20 percentage points above exact indexing and source reads fall by 73.33%, at the cost of 12 additional reads per seed; the state gate pays off only on bad claims that have not yet completed a second independent practice (0 errors versus 209 errors). These results support experience reuse within bounded contractual tasks; they do not demonstrate self-evolution of model parameters or general improvements in language-model capability.

Index Terms— Practical experience; experience lifecycle; evidence constraints; version revision; host agent; Bayesian evidence updating.

Author and Acknowledgments

Wei Xu holds a Bachelor of Engineering degree in Electronic Science and Technology. This research was conducted independently after graduation. The author thanks Yancheng Institute of Technology for the laboratory environment, coursework, and knowledge acquired during his studies. This statement describes educational background and acknowledgment only; it does not identify the institute as the author's affiliation or the institution conducting this research.

I. Introduction: Experience as Knowledge Tested through Practice

When addressing a problem, a person generally begins with a goal and acts using existing knowledge. Action produces an outcome, and comparison with the expected outcome yields feedback. Feedback does not automatically become experience: the outcome may be accidental, conditions may be undocumented, observations may be misinterpreted, or a formerly successful method may fail in a different environment version. The essential property of experience is the organization of action outcomes into knowledge that can be reused under explicit conditions, challenged, and revised. Agent-memory systems often emphasize storage and retrieval while leaving the validity, expiration, and correction of remembered knowledge less explicit [1-2].

This paper studies computational modeling of experience acquired through practice. The system, Butian, runs within a host AI application. The host supplies the model, sessions, and tools; Butian manages experience objects, evidence chains, version admission, state transitions, feedback persistence, and context recovery. Software memory is not equated with human consciousness, and textual records are not equated with parameter learning. Human practical reasoning is abstracted into an observable engineering process, whose usefulness is assessed through states, evidence, recurring errors, reuse benefits, and recovery quality.

A. Research Questions and Contributions

RQ1 asks whether the action–outcome–reflection–validation–reuse process can be represented as a state machine preserving evidence and conditions. RQ2 asks when candidate experience may become stable and when the system must return unknown or record a dispute. RQ3 asks whether the mechanism reduces repeated evidence acquisition and avoids obsolete-rule errors under version changes, source outages, and service restarts. RQ4 concerns the distinct roles of Bayesian updating and exploration strategies. The contributions are a layered definition of practical experience, algorithms from evidence production to revision, multi-round mechanism experiments using the real ButianService, independent numerical controls, and explicit boundaries between host-AI evidence and mechanism experiments.

B. Related Work and Positioning

Experiential-learning research connects concrete experience, reflection, conceptualization, and subsequent experimentation. The systematic review of Kolb's model emphasizes context and reflection [3]. Accordingly, we distinguish having performed an action from having formed reusable experience: an individual occurrence is an event, an explanation is a candidate claim, and subsequent practice supplies further support or counterexamples. This relationship informs the research design; it does not imply a software threshold of two successes. That threshold is an auditable engineering choice requiring sensitivity analysis in specific task families.

Agent research provides complementary references. Reflexion converts task feedback into verbal reflection, and Voyager supports subsequent exploration through reusable skills [4-5]. MemoryBank studies long-term memory, while LongMemEval evaluates cross-session memory [1-2]. These studies do not independently establish the correctness of our thresholds or version rules. Here, provenance, nonduplicated evidence, renewed verification after version changes, and preserved disputes are placed within one practical process and

examined separately. Retrieval-augmented generation provides access to external information [6], but retrieving relevant text still leaves an applicability decision to be made.

C. Organization

Section II defines the events within a practical episode. Section III explains candidate experience, evidence, counterexamples, versions, and reuse admission. Section IV presents algorithms and implementation. Sections V and VI describe experiments and results. Section VII discusses the roles of Bayesian components, engineering benefits, and limitations. Section VIII concludes. Mathematical tools support, rather than replace, the central account of experience formation.

II. A Process Model of Experience Formation through Practice

A. From Goals to Action Outcomes

At time t, let a person face goal G_t, environment E_t, and existing knowledge M_t. The person proposes action A_t, changes the environment through tools or physical action, and obtains observation O_t. The difference between the expected observation $\hat{O}_t$ and actual observation produces feedback $F_t = O_t - \hat{O}_t$. This is not merely a correct/incorrect label. An unexecuted action has no interpretable outcome; success cannot be transferred when the environment version is unknown; observations without provenance cannot be audited; and one occurrence does not establish statistical stability. The model therefore retains actions, environment snapshots, observation provenance, and outcome evidence rather than a summary alone.

$$(G_t, E_t, M_t, A_t) \rightarrow O_t \rightarrow F_t \tag{1}$$

B. From Feedback to Candidate Knowledge

Reflection asks what happened, why it happened, and under which conditions it may be reused. Candidate experience e contains claim h, applicability conditions c, environment version v, provenance s, evidence set E, and counterexample set X. A candidate is a testable interpretation, not an established fact. Outcomes without conditions create overly broad candidates; interpretations without provenance cannot be audited; deleting counterexamples obscures applicability boundaries.

$$e = (h, c, v, s, E, X, l, \theta) \tag{2}$$

Here l is lifecycle state, and θ summarizes evidence-supported belief. It is an estimate under the current evidence and working prior, not the probability of truth. A candidate may progress to warm only after further successful predictions in practice and in the absence of unresolved counterexamples.

C. Renewed Practice and Revision

A subsequent task first checks whether its conditions satisfy c and v. After adoption, the first-action outcome is frozen so that later answers cannot contaminate first-attempt evaluation. A successful prediction adds an observation with an independent key and evidence reference. Failure adds a counterexample and initiates dispute or revision. When sources, tools, or termination status cannot be confirmed, the result is unknown. Revision creates a new version with its parent, reason for change, and evidence rather than overwriting the earlier account.

D. Running Example: Formation of a Calibration Rule

Consider a sensor using a linear conversion in version v1. The first task specifies the entity, software version, and raw measurement. The host selects a rule from source material and produces an output; an instrument-calibration record supplies outcome evidence. Instead of recording a universal rule, the system creates the candidate that rule h is effective for entity a, version v1, and the stated measurement range. A second success with a different calibration sample and evidence digest satisfies the local threshold. Different files may nevertheless originate from the same erroneous source: deduplication removes recognized duplicates but cannot guarantee statistical independence.

After an update to v2, the old conditions no longer cover the task. The system should seek v2 evidence and retain unknown if that evidence is unavailable. Once sources recover, it creates a revision and tests it in further practice. Failure under genuinely matching v1 conditions attaches a counterexample to the original claim and changes its state to disputed; a more confident explanation cannot erase that failure. This example connects goals, actions, evidence, candidates, reuse, drift, and revision as one process.

E. Five Essential Stages

The process requires goal specification, action, observable outcomes, validation, and renewed practice. Without a target problem, experience has no task of application. It must constrain actual action, retain outcome evidence, check conditions and counterexamples, and be tested again in later tasks. Butian's state machine, evidence tables, version tables, recovery mechanism, and feedback interfaces implement observable counterparts of these stages. Omitting a stage leaves a log, a retrieval result, or a candidate claim rather than a complete experience process.

III. Experience Lifecycle: From Candidates to Reusable Knowledge

A. States

candidate denotes experience from one or a few episodes that has not met stable-use conditions. warm denotes experience meeting the specified success and evidence thresholds and eligible for retrieval under matching conditions. hot denotes recently effective and frequently used experience; frequency is not additional proof of correctness. disputed records counterexamples, conflicts, or environmental mismatch and is not unconditionally reusable. invalidated records a claim known to no longer hold. unknown describes an unresolved observation or task outcome, not a memory-temperature state alongside candidate. The implementation also includes maintenance states such as cold, archived, and pending_validation. These names describe program records, not psychological states.

$$s \geq 2, f = 0\text{: candidate} \rightarrow \text{warm} \tag{3}$$

$$\text{failure: warm} \rightarrow \text{disputed} \tag{4}$$

B. Entry Requirements

External articles and host responses can supply candidate sources but cannot directly establish stable local experience. Entry requires a claim, applicable task, source event,

environment or version information, and a traceable outcome. A success must include evidence_ref. The independent_key, evidence digest, and optional content hash support deduplication. Repeated use of one file cannot masquerade as independent practice.

C. Reuse Admission

Admission identifies the task entity and target version, retrieves candidates, and then checks uniqueness, content completeness, conditions, and parent-chain continuity. Conflicts or budget truncation cause fallback to the exact source; unavailable sources yield unknown. This models recalling context and comparing conditions before using prior experience. Production compose_context supplies candidates only. The research parser and host perform final admission, so retrieval itself must not be described as understanding.

D. Effects of Failure

Failure does not silently delete experience. Its evidence, environment, version, and first answer are retained. Failure of mature experience produces disputed; failure of a candidate remains in its observations and blocks unconditional promotion. Narrowed applicability creates a revision, whereas a wholly unsupported claim can become invalidated. The record preserves why the claim was previously accepted and why it is now revised.

IV. Algorithms and Implementation

A. Algorithm 1: Practice Recording and Candidate Creation

Inputs are goal G, action A, environment snapshot E, observation O, provenance s, and expectation $\hat{O}$. An unfinished action or missing evidence is recorded as unknown and cannot trigger promotion. Success creates or retrieves an experience object and checks independent keys, evidence digests, and content hashes; duplicate evidence is rejected. Failure records a counterexample and changes mature experience to disputed or retains a candidate. Observations, evidence digests, experience versions, and domain-skill updates commit within one BEGIN IMMEDIATE transaction in MemoryStore. ButianService subsequently appends practice_observed to a separate service store. These operations are not a cross-database atomic commit.

$$r_t \in \{\text{success}, \text{failure}, \text{unknown}\} \tag{5}$$

B. Algorithm 2: Beta Updating of Experience Belief

Under the working model that a claim succeeds on the next condition-matching task, let $\theta \sim \text{Beta}(\alpha, \beta)$. Evidence weight is $w = qdm$, where q is evidence quality, d an independence assessment, and m environment match. Success yields $\alpha' = \alpha + w, \beta' = \beta$; failure yields $\alpha' = \alpha, \beta' = \beta + w$. The posterior mean is $\mu = \alpha/(\alpha + \beta)$. Its increase after success is $\Delta\mu = w\beta/[(\alpha + \beta)(\alpha + \beta + w)]$. Thus success increases working belief with diminishing marginal change as parameters grow. This characterizes evidence accumulation without proving the claim.

$$\theta \sim \text{Beta}(\alpha, \beta), \mu = \frac{\alpha}{\alpha+\beta} \tag{6}$$

Production record_practice_outcome currently uses bounded adjustments of confidence: failure subtracts 0.1, and qualifying candidate promotion adds 0.1. It does not automatically call BetaBelief. The Beta model is an optional representation supported by the mathematical module and controls; production confidence is not a Beta posterior. Fractional weighting approximates a weighted likelihood and does not solve evidence duplication. Independence assessment requires higher-level deduplication and environmental review. A posterior cannot bypass version admission or practice thresholds, and warm does not mean proved.

C. Algorithm 3: Candidate Selection and Information Gain

For actions that may reduce uncertainty, information gain is $\text{IG} = H(p_{\text{prior}}) - H(p_{\text{posterior}})$; a cost-adjusted score is $\text{IG} - \lambda\text{cost}$. Expected information gain has a mutual-information interpretation only when the posterior distributions and outcome probabilities satisfy normalization and mixture consistency. Butian's expected_information_gain checks explicit posterior dimensions, outcome probabilities, and mixture consistency. It does not infer posteriors automatically from prose.

$$\text{EIG} = H(p) - \sum_y r_y H(p(\cdot\,|y)) \tag{7}$$

D. Algorithm 4: The Role of Thompson Sampling

Thompson sampling allocates exploration among actions or strategies awaiting verification; it is not the experience-formation process. Arm k maintains $\text{Beta}(\alpha_k, \beta_k)$. Each round samples $\tilde{\theta}_k$, selects the largest sample, and updates only the selected arm using Bernoulli feedback. Selection costs $O(K)$ and updating costs $O(1)$. Without state transitions, long-term returns, or value functions, the current component is a context-free Bayesian bandit rather than full Bayesian reinforcement learning.

E. Algorithm 5: Recovery after Restart

Recovery reads committed events, current experience versions, parent chains, and unfinished tasks before reconstructing candidate context. A prepared event without committed status cannot establish successful external action. Unconfirmed tool termination remains unknown. A version mismatch permits only candidate use and renewed evidence acquisition. Recovery continues the practical process rather than merely reinserting old text into context.

F. Pseudocode with Inputs, Outputs, and Error Branches

The responsibility boundaries below are explicit. Algorithm 1a is the production practice interface; Algorithms 2a–4a are mathematical modules. Algorithm 5a describes host-side recovery and admission rather than a single implemented function. Algorithm 6 is the actual main-experiment path and writes warm directly; it does not replace candidate validation through Algorithm 1a. Numbers indicate execution order; blocks terminate with END.

Algorithm 1a. Record Practice Feedback (Production Interface)

text 01 INPUT: member, memory_id, content, outcome, independent_key, evidence_ref 02 OUTPUT: memory_id, status, successes, failures; or an explicit error 03 REQUIRE outcome IN {success, failure, unknown}; independent_key nonempty 04 IF success AND no evidence_ref: RAISE practice_success_evidence_required 05 Service checks evidence ownership and content hash; RAISE if invalid 06

BEGIN IMMEDIATE (member-memory transaction) 07 IF independent key or evidence reference/digest already used: RAISE duplicate 08 Read experience; create candidate if absent; check requested content hash 09 Insert observation, environment, digest; count successes s and failures f 10 IF failure: mature -> disputed; otherwise retain candidate; reduce scores 11 ELSE IF success AND s>=2 AND f=0 AND old state IN {candidate,pending_validation}: set warm and increase bounded scores 12 ELSE: no promotion; unknown contributes neither success nor failure 13 Save revision, parent chain, provenance, domain statistics; COMMIT 14 ON transaction exception: ROLLBACK; rethrow 15 Append practice_observed separately; do not claim two-store rollback 16 RETURN actually committed state and counts

Algorithm 2a. Beta Evidence Summary (Optional Mathematical Module)

text 01 INPUT: alpha>0, beta>0, outcome, q,d,m IN [0,1] 02 OUTPUT: updated parameters and mean; production lifecycle unchanged 03 Check finite parameters and valid ranges; RAISE if invalid 04 w <- q*d*m 05 IF success: alpha <- alpha+w 06 ELSE IF failure: beta <- beta+w 07 ELSE IF unknown: leave parameters unchanged 08 ELSE: RAISE invalid_outcome 09 RETURN alpha, beta, alpha/(alpha+beta) 10 NOTE: Algorithm 1a deduplicates; fractional weight proves no independence

Algorithm 3a. Rank Verification Actions (Explicit Posterior Control)

```text 01 INPUT: prior p; outcome probabilities r; posteriors p_y; cost; lambda 02 OUTPUT: EIG and cost-adjusted scores for valid actions 03 FOR EACH action a:
04 Check finite nonnegative p,r,p_y; matching dimensions; normalization
05 Check sum_y r_y*p_y[i] matches p[i] within tolerance for each i
06 RAISE if invalid; do not guess posteriors from text
07 eig[a] <- H(p)-sum_y r_y*H(p_y)
08 score[a] <- eig[a]-lambda*cost[a]
09 END FOR 10 RETURN score-ranked actions; the caller decides execution ```

### Algorithm 4a. Thompson Exploration Allocation (Bandit Control)

text 01 INPUT: positive alpha[k],beta[k] for K arms; random-number generator 02 OUTPUT: selected arm; its parameters updated after feedback 03 REQUIRE K>0; finite positive parameters 04 FOR k=1 TO K: sample[k] <- BetaSample(alpha[k],beta[k]) 05 END FOR 06 selected <- ARGMAX sample[k], with implementation-defined tie handling 07 Execute selected; await verifiable Bernoulli reward 08 IF unknown: fabricate no reward and perform no update 09 ELSE IF reward=1: alpha[selected] <- alpha[selected]+1 10 ELSE IF reward=0: beta[selected] <- beta[selected]+1 11 ELSE: RAISE invalid_reward 12 RETURN selection and parameters; no automatic experience promotion

### Algorithm 5a. Conditional Admission after Recovery (Host Workflow)

```text 01 INPUT: persisted tasks, events, experience, chains, current conditions 02 OUTPUT: auditable candidate context and outcome status; no automatic success 03 Open service and member stores; read current versions and provenance 04 Check parent chains; report corruption without inventing missing versions 05 FOR EACH unfinished action:
06 Recover committed outcomes only with verifiable evidence
07 Retain unknown for uncommitted prepared actions, unconfirmed exit,
or missing evidence 08 END FOR 09 Retrieve candidates; admit by entity, version, completeness, and conflict 10 IF inapplicable: reacquire source; IF unavailable: RETURN unknown 11 RETURN candidates, sources, unresolved matters; host must practice again 12 NOTE: experiments test normal close/open, not arbitrary power failures ```

### Algorithm 6. Multi-round Rule-Reuse Benchmark (Executed Research Path)

text 01 INPUT: shared task stream, exact source archive, method, memory directory 02 OUTPUT: first answers; correct/wrong/unknown; read/query/write counts 03 IF memory enabled: compose_context; parse a unique entity rule 04 IF version admission enabled: reject mismatch; ELSE permit old rules 05 IF no usable rule AND source_available: scan or read exact index 06 answer <- fixed solve(rule,task); freeze first answer 07 Score against reference; later correction cannot change first score 08 IF first answer wrong AND source_available: reacquire correct rule 09 IF source acquired AND writes allowed: append_event; MemoryStore.put(status=warm,confidence=1) 10 At round 8 close/open; no-persist switches to an empty directory 11 RETURN round record; repeat for subsequent rounds 12 NOTE: no candidate validation, automatic induction, or disputed-state study

## G. Dream Consolidation, Hypothesis Verification, and Subsequent Practice

Dreaming is offline organization of existing practice material. DreamRunner.run creates a source snapshot when material volume, new-material, interval, and budget thresholds are met. The host supplies distilled summaries, abstractions, and hypotheses; the system does not derive rules automatically from text. distill preserves source references, abstract retains support, counterexamples, and conditions, and propose checks continuing source validity. Verification must pass schedule and start before supported or rejected outcomes are accepted, and evidence is checked by the service. supported creates only a candidate; later successful practice is still required for warm. Verification separates testable hypotheses from unchecked generalization without guaranteeing correctness on arbitrary tasks.

### Algorithm 7. Dream Formation and Reuse after Verification

text 01 INPUT: practice material, host-proposed rule, verification budget, later tasks 02 OUTPUT: supported/rejected/unknown; candidate; later practice records 03 Check new material, interval, budget; skip consolidation if unmet 04 Freeze sources; host distills, abstracts conditions, and proposes hypothesis 05 Schedule verification; start only if authorized and sources remain valid 06 Execute probes distinct from training; save observations and evidence 07 IF counterexample: record rejected; use source fallback on later tasks 08 ELSE IF insufficient evidence: record unknown; do not claim validation 09 ELSE: record supported and create candidate 10 Freeze first outcomes and record independent practice feedback during reuse 11 Promote to warm only at production threshold; preserve failure evidence 12 RETURN verification/reuse ledger; supported does not mean universal validity

### H. Conditional Propositions for the Bounded Loop

Proposition 1: If every state update uses the MemoryStore practice interface, candidate promotion requires at least two success observations passing interface deduplication and no failures. The promotion branch establishes this by checking counts transactionally and requiring the prior state to be candidate or pending_validation. This proves a program gate, not external evidence truth, and it does not cover manual writes bypassing the interface. Two successes are necessary at this interface, not sufficient for general validity.

Proposition 2: If the current task version is identifiable and the host strictly enforces version admission, an old-version candidate cannot directly yield an accepted current-version result. Matching candidates proceed normally; mismatches require new evidence or unknown. The claim depends on actual version identification and enforcement. The no-version ablation removes that assumption; its errors demonstrate the guard's role rather than universal immunity to errors.

Proposition 3: An unknown observation contributes neither a success nor a failure. Counting includes only the corresponding explicit outcomes. Unknown observations can retain records and versions but cannot increase support. This prevents missing sources or unconfirmed termination from being interpreted as success. Unknown outcomes still reduce completion and remain in the evaluation denominator; they are the cost of honestly representing uncertainty.

A complete process also requires host execution, reliable observations, and later reuse. Database success establishes persistence only. Validity across tasks requires external outcomes, and long-term benefits require cross-round controls. Executable closure and epistemic correctness are therefore argued separately: interfaces, transactions, and recovery controls support the former; the stated task families bound the latter.

### I. Complexity and Implementation Boundaries

Practice writes involve a fixed set of table-operation types, but success/failure counting and domain aggregation grow with observations. For $N$ relevant observations and $N_d$ domain observations, conservative aggregation cost is $O(N + N_d)$ without stronger index guarantees. The entire feedback operation is not $O(1)$, and a fixed number of SQL statements does not make their runtime constant. Current OR-LIKE retrieval costs conservatively $O(ML)$ for $M$ experiences of average content length $L$. Candidate parsing costs $O(B)$ for $B$ context characters; version-chain checking costs $O(V)$; Beta updating is $O(1)$; Thompson selection is $O(K)$. Pareto budget allocation depends on frontier size, capped at 100000. These are program costs, not host-token costs or human cognitive complexity.

## V. Experimental Design

### A. Main Experiment and Scope

The main experiment is a rule-reuse benchmark, not a test of the full experience-formation process. Its remember function calls MemoryStore.put directly with warm and confidence 1.0, bypassing candidate thresholds, independent practice validation, and disputed-state handling. A generator supplies entities and versions; a fixed solver applies rules; an evaluator compares references; source rules are then cached for later reuse. No language model is called. The experiment measures infrastructure supporting a practical cycle rather than language generation. Ten seeds, 12 entities, 12 rounds, and seven conditions produce 10080 executions. Version drift occurs in rounds 5 and 9, source outages in rounds 6 and 7, and normal service close/reopen in round 8.

### B. Conditions and Controls

B0-scan scans the archive every round; B0-index uses an exact version index. B1-static writes only in the first round. B2-full enables retrieval, version checking, source fallback, feedback writes, and persistence. no-version removes admission, no-write disables experience writes, and no-persist uses an empty directory after restart. First outcomes are frozen before feedback, and correct, wrong, and unknown are mutually exclusive. The statistical units are the 10 seeds, not repeated rounds treated as independent samples.

### C. Metrics and Hypotheses

Primary measures are first-attempt completion, version errors, unknown outcomes, and source reads. Secondary measures include context queries, writes, logical operations, local runtime, and version-chain audits. H1 predicts fewer repeated source reads and avoidance of obsolete rules under drift and outages with version admission. H2 predicts retained reusable experience after restart. H3 allows increased local overhead, so fewer source reads need not imply lower end-to-end latency.

### D. Lifecycle Interface Control and Limitations

A separate 24-directory control uses record_practice_outcome: first success creates a candidate; the same evidence with a new independent key is rejected; a second distinct evidence item promotes to warm; failure produces disputed; the script manually revises to candidate; normal close/open is followed by checks of content, parent chains, and retrieval. This verifies state guards, not autonomous host induction, true evidence independence, or claim correctness. Evidence and revisions are scripted. Twenty-four repetitions are not 24 scientific discoveries. This control complements rule reuse but was originally not one end-to-end experiment jointly measuring formation and task benefit; Section V.G closes this gap.

### E. Paired Replay of Dream Effects

Ten seeds generate linear and saturating sensors, yielding 20 material groups. Initial observations are at x=1,2,3. The current host proposes an affine candidate based on equal adjacent differences; Python replays this fixed induction template through production dream interfaces. The no-dream condition reads sources for each problem. Unverified reuse adopts the candidate directly. Verified reuse probes x=5,10, rejects and falls back on a counterexample, or saves a candidate and records subsequent feedback. Tests at x=4,7,9,12 repeat over four rounds, giving 320 tasks per condition. Probe and test inputs differ. Each condition has identical initial material and 60 training reads. Saturating sensors plateau above x=6, so the initial three points cannot distinguish the families. This examines overgeneralization and verification cost; it is not a blinded autonomous-discovery study and supplies no independent model samples.

### F. Independent Algorithm Controls

A separate two-arm Bernoulli experiment runs 10 fixed seeds and 1000 rounds per seed with reward probabilities 0.30 and 0.70, comparing Thompson sampling, epsilon-greedy, and

uniform random selection. It tests exploration allocation only, has actual_model_runs=0, and is neither the main practical experiment nor a full Bayesian-RL study.

### G. End-to-End Gate Experiment Design

To connect formation and task benefit within a single experiment, we add an end-to-end gate benchmark (script run_endtoend_gate_study.py; the criteria were written into a protocol and frozen before the run). It reproduces the main study's task stream statement for statement, so tasks, drift, outages, and restart are identical; the only variation is how experience is admitted as reusable. Six conditions: G0 exact indexing; G1 writes a source rule as warm the first time it is fetched, which is the B2-full write path of Table 1; G2 records each source read as one independent practice and requires two mutually distinct evidence items before promotion; G3 adds, on top of G2, a bad claim whose state is candidate; G4 equals G3 except that candidate-state claims are also treated as reusable; G5 adds, on top of G2, a bad claim carrying two independent practice records. Thus G3 minus G4 isolates the state gate, G3 minus G5 isolates the second independent practice, G2 minus G1 measures the total cost of the gate, and G2 minus G0 measures the benefit of experience against no experience.

Bad claims are injected before rounds 6 and 10 into the entity that drifted in the preceding round: the content is the pre-drift old rule and the version field is rewritten to the current version, so version and content are internally consistent while the content is stale. This is the most common production form of a bad claim and the only external shock the experiment imposes. Twenty seeds (the original ten plus ten new), 12 rounds, and 12 entities give 17,280 executions. The replication self-check requires G1 to match the published B2-full row by row on the ten original seeds and repeated runs on the same machine to match field by field; the statistical unit is the seed and intervals are seed-cluster bootstraps.

## VI. Results: Benefits and Costs of Experience Reuse

### A. Main Results

Table 1. First-attempt outcomes and evidence-acquisition costs.

| Condition | First-attempt completion | Wrong | Unknown | Source reads |
|---|---|---|---|---|
| B0-scan | 1375/1440 | 0 | 65 | 42287 |
| B0-index | 1375/1440 | 0 | 65 | 1375 |
| B1-static | 1400/1440 | 0 | 40 | 752 |
| B2-full | 1440/1440 | 0 | 0 | 253 |
| no-version | 1307/1440 | 133 | 0 | — |
| no-write | 1375/1440 | 0 | 65 | 1375 |
| no-persist | 1440/1440 | 0 | 0 | 373 |

B2-full names all switches of the rule-reuse policy, not the full experience lifecycle. Relative to exact indexing, completion improves by 4.51 percentage points, with a seed-cluster bootstrap 95% interval of [3.68,5.35]. Reads fall from 1375 to 253, an 81.60% reduction. Removing version checks produces 133 obsolete-rule errors: version is an applicability condition, not an incidental field. Discarding experience on restart adds 120 reads, showing that persistence changes subsequent evidence-acquisition actions.

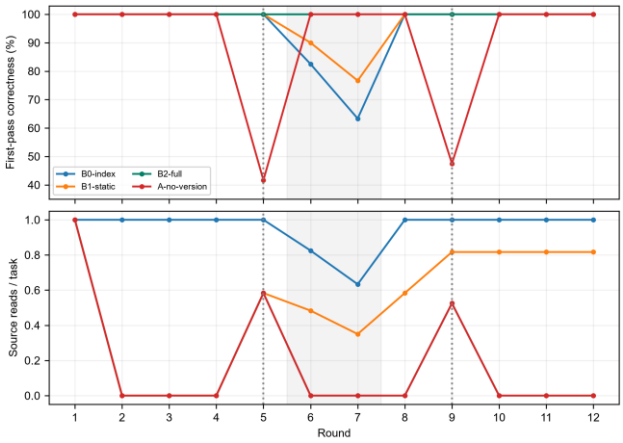



Figure 1. Outcomes under round-specific drift, outages, and restart.

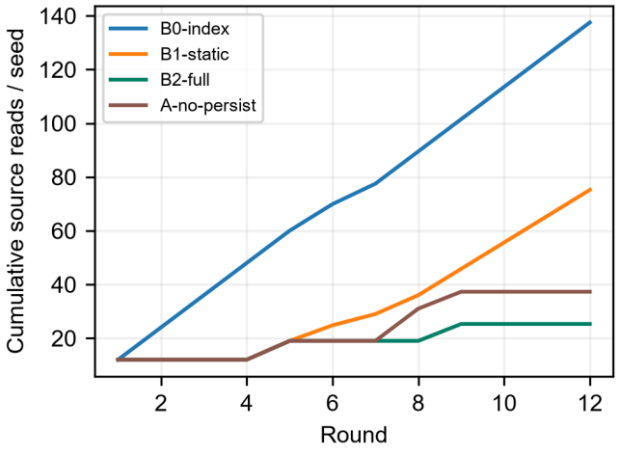



Figure 2. Effect of reuse on cumulative source access.

### B. Round-by-round Attribution

Table 2a. First outcomes by round and B2 external-source reads.

| Round | Index correct/120 | B2 correct/120 | B2 reads |
|---|---|---|---|
| 1 | 120 | 120 | 120 |
| 2–4 (each) | 120 | 120 | 0 |
| 5 (drift) | 120 | 120 | 70 |
| 6 (outage) | 99 | 120 | 0 |
| 7 (outage) | 76 | 120 | 0 |
| 8 (restart) | 120 | 120 | 0 |
| 9 (drift) | 120 | 120 | 63 |
| 10–12 (each) | 120 | 120 | 0 |

Both policies read 120 sources in round 1, providing no cold-start advantage. In rounds 2–4, B2 reuses unchanged rules while indexing still reads 120 sources per round; accuracy is identical. Round 5 updates 70 entity instances: admission rejects 70 old rules and reacquires sources while retaining the other 50. Round 9 requires 63 new reads. Thus total B2 reads decompose exactly as 120+70+63=253; 1122 avoided reads arise from subsequent repeated tasks.

All completion differences occur in rounds 6 and 7. The index returns 21 and 44 unknown outcomes because sources are unavailable; B2 solves these 65 tasks using previously saved version-matching rules. Both solve all 1375 available-source tasks. Among the 65 unavailable-source tasks, index completion is 0/65 and B2 completion 65/65. The 4.51-point gain is therefore an information-availability benefit under this protocol, not stronger reasoning, formulas, or language understanding. Updating rules in round 5 before outages gives B2 a learning opportunity; simultaneous first-time version changes and outages are not covered.

Without admission, 70 round-5 and 63 round-9 tasks use obsolete rules, totaling 133 first-attempt errors. Feedback occurs after freezing the first answer. Available sources then correct the cache, preventing propagation into later stable rounds. After normal restart, B2 has 120 hits, whereas the empty-directory policy rereads 120 sources. Both solve 120/120. Here persistence avoids rereading; it does not improve round-8 accuracy or guarantee power-failure recovery.

### C. Operation Costs and Boundaries

Table 2. Operation ledger for experience formation and reuse.

| Condition | Reads R | Queries Q | Writes W | Logical operations |
|---|---|---|---|---|
| Exact index | 1375 | 0 | 0 | 1375 |
| B2-full | 253 | 1440 | 253 | 1946 |
| B1-static | 752 | 1440 | 120 | 2312 |
| no-persist | 373 | 1440 | 373 | 2186 |

B2-full increases logical operations by 41.53%, averaging 8.44 ms locally versus approximately 0.0102 ms for indexing. The mechanism reduces source acquisition and obsolete-rule errors, not necessarily latency. This local benchmark does not measure human learning cost. With equal operation weights, reuse has no total-operation advantage. Savings may outweigh management overhead only when external reads are sufficiently more expensive than local retrieval and writes. External fees, network latency, and model tokens were not measured, so an application break-even point cannot be established.

### D. Independent Bayesian Bandit Control

Table 3. Exploration-allocation control.

| Policy | Mean total reward | Mean pseudo-regret | Seeds |
|---|---|---|---|
| Thompson | 704.4 | 4.32 | 10 |
| epsilon-greedy | 672.8 | 25.44 | 10 |
| Uniform | 494.3 | 202.60 | 10 |

Thompson sampling concentrates relatively quickly on the higher-reward arm in this stationary two-arm Bernoulli environment. This describes the allocator, not the correctness of experience claims or reinforcement-learning capability in the host model.

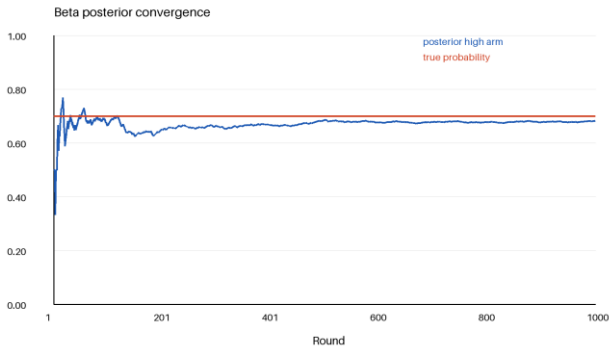



Figure 3. Convergence of the high-reward candidate's Beta posterior mean.

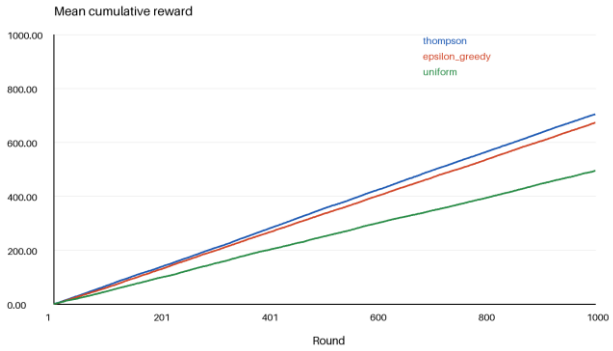



Figure 4. Mean cumulative reward of the three exploration policies.

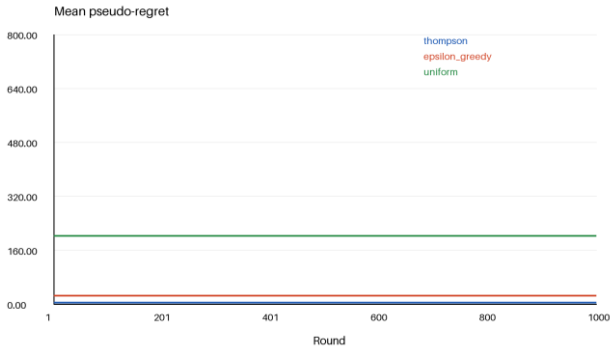



Figure 5. Mean pseudo-regret of the three policies.

## E. Subsequent Dream Benefits and Incorrect Generalization

Table 4. Paired outcomes for candidate verification and later reuse.

| Condition | Correct/320 | Wrong | Later source reads | Verification accesses |
|---|---|---|---|---|
| No dreaming | 320 | 0 | 320 | 0 |
| Unverified reuse | 200 | 120 | 0 | 0 |
| Verified reuse | 320 | 0 | 160 | 40 |

Verification accepts 10 linear candidates and rejects 10 saturating candidates. Linear rules solve 160 later tasks by reuse; the other 160 use fallback. Later reads fall 50%, from 320 to 160. Including 40 verification accesses gives 200, a 37.5% reduction. Including the identical 60 training accesses per condition gives 380 versus 260, a 31.58% reduction. Unverified reuse fails on three above-threshold inputs per saturating sensor per round: 10×4×3=120 errors. Compression into a rule can lose boundaries. Verified reuse matches no-dream accuracy; its benefit is reduced acquisition rather than improved reasoning accuracy.

Accepted candidates pass through the real record_practice_outcome: the first success retains candidate, and the second promotes to warm. This connects dream candidates, tool verification, and later practice. However, reference outcomes come from the same synthetic generator and remain visible to the evaluator; they are not free external evidence in deployment. The induction template was specified by the host, with zero model calls during script execution. Repeated rounds within a seed are not independent samples. The results establish verification's role in this protocol, not the success rate, token cost, or cross-domain generalization of unconstrained host induction.

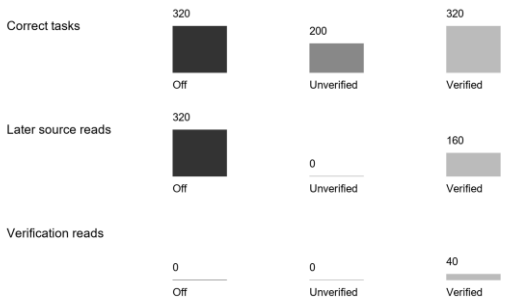



Figure 6. Correct tasks, later source reads, and verification accesses under the three dream conditions.

## F. End-to-End Gate Experiment: Linking Formation to Task Benefit

Table 5. First-attempt outcomes and gate costs of the six conditions (20 seeds pooled, 2880 tasks per condition).

| Condition | Correct | Errors | Unknown | Source reads | Writes | Gate rejections |
|---|---|---|---|---|---|---|
| G0 exact index | 2759 | 0 | 121 | 2759 | 0 | 0 |
| G1 direct write to warm | 2880 | 0 | 0 | 496 | 496 | 0 |
| G2 production gate | 2880 | 0 | 0 | 736 | 736 | 240 |
| G3 gate plus candidate bad claim | 2844 | 0 | 36 | 992 | 992 | 449 |
| G4 no gate, candidates reusable | 2653 | 209 | 18 | 752 | 943 | 0 |
| G5 gate plus two-evidence bad claim | 2634 | 209 | 37 | 1601 | 1792 | 868 |

The replication self-check yields two results: on the ten original seeds G1 totals 1440 correct answers and 253 source reads, identical row by row to B2-full in Table 1; and the task stream matches the published archive item by item, while a repeated run on the same machine matches all 864 rows field by field. The 4.51 percentage points of Table 1 are therefore obtained under the shortcut of writing a fetched source rule to warm immediately, which bypasses the candidate gate and the second independent practice; with the gate returned to the loop, the gain of G2 over G0 is 4.20 percentage points (6.05 additional solved tasks per seed, seed-cluster bootstrap 95% interval [3.65, 4.79] percentage points, positive for all 20 seeds). The two numbers are of similar magnitude but different meaning: the former is an upper bound describing the situation in which experience already exists, the latter includes the cost of the gate.

The cost of the gate is an identity: G2 reads 12.0 more sources per seed than G1 (interval [12, 12]), exactly equal to its 12.0 gate rejections per seed, while accuracy is identical (all 20 seeds tied). The gate therefore removes no accuracy; it exchanges each hit that the state does not allow for one source read. Against the experience-free index baseline, G2 reads 2023 fewer sources and solves 121 more tasks, a 73.33% reduction in reads.

The effect of the state gate narrows to one provable point. The same bad claim (256 injections) produces 0 first-attempt errors on G3, 209 errors on G4 with all 256 items fooling the system at least once (median 1 per item), and exactly the same error count on G5 as on G4. Adding the second independent practice therefore does not reduce how often a bad claim deceives: the gate blocks claims that carry only a single evidence impression and have not completed a second independent practice, not claims that carry evidence. The cost of the gate appears as coverage: the 36 unknown outcomes of G3 all fall in rounds 6 and 7, where the gate rejects and no source is available, so a possible wrong answer becomes a missing one.

The feedback channel is one-directional. The probe shows that a warm memory enters disputed after one failure and does not return to warm after two further successes; only an explicit revision reopens it. In the main study all 191 hit-then-wrong-then-corrected events fall inside this one-directional interval (G4 lands in candidate, G5 in disputed). This is not an implementation error but a direct consequence of the promotion rule requiring a zero failure count, and we record it as an open limitation.

## G. Interpretation

The main study supports condition-matching reuse of known rules, not autonomous formation of claims from practice. B1 caches only the first round and can still read sources after updates; its limitation is repeated acquisition, not inevitable obsolete outputs. Formation, thresholds, and disputes were originally covered only by the interface control; the end-to-end gate experiment of Section VI.F links them to task benefit in one ledger, and the conclusion is that the gate is real but narrow: it blocks claims that have not completed a second independent practice, it does not raise accuracy, and it does not block claims carrying two evidence items. Bandit benefits arise from exploration allocation at a different level. These data cannot be merged into one model-effect number.

## H. Real-Host Multiround Experiment

The preceding three experiments were hosted either by a deterministic solver or by a single-round, non-blinded pilot. To add a real model driving real tools over multiple rounds with objective scoring, a pre-registered protocol was written before any host answer: the model of the current session calls the recall and source-read tools directly over JSON-RPC on standard input and output, with only three allowed actions — recall experience, read the source, and submit an answer. Experience writeback is performed by the harness after exact-match scoring, so the host has no channel for writing experience. Twelve entities are arranged as 3 families x 4; the first two of each family keep one version across all three rounds and the last two are upgraded each round, giving 36 answers in total. This is a pilot and no p-values are reported.

All five pre-registered criteria hold. Stable entities score 6/6 in the first and in the last round, so accuracy does not

degrade; drifting entities produce zero obsolete-rule errors in the last two rounds, while each of those rounds reads the source exactly six times, equal to the number of drifting entities; stable entities drop to zero source reads in the last two rounds without losing correct answers; at the last round the six stable memories are warm (three independent practices each, under three independent practice keys), while the eighteen drifting memories split into three per-version entries that all stay candidate; and all 36 answers carry a recall record, with every reuse answer naming a memory that appears in that round's recall payload. Source reads total 24, a 33.3% reduction against the 36 reads that reading for every entity in every round would require, and the saving falls entirely on the stable entities in the last two rounds.

Three limitations must be stated together. First, accuracy is saturated (36/36), so "no degradation" holds as a tie rather than as an improvement and the informative content is behavioural: version admission was executed by the host itself after comparing the version field inside the recalled content, and a host that lazily reused the stale rule would have made six more errors in each of the last two rounds. Second, the host is non-blinded and the arithmetic of each question was executed by a local script; the experiment shows that the channel, version-admission, and reuse chain works when driven by a real host, not that model reasoning improved, and it does not generalize across models. Third, two construction defects that would have made the questions unanswerable were fixed before any answer was produced: rules were originally generated per round, so the archive carried three different RULE lines for the same entity and version, and the archive did not state the conversion formula. With those fixed, a failed criterion was preferred to a claim built on an unanswerable apparatus. The experiment also exposed an interface shortcoming: recall performs substring matching only, with no structured filter by entity or tag, so with limit = 10 only ten of twelve memories are returned and the queried entity itself can be the one missing; the host must raise the limit and filter by itself. This is recorded as an interface defect to fix, not as a mechanism result.

## VII. Discussion: From Human Practice to Host-Agent Mechanisms

### A. What Is Modeled

The model represents the structure of practice rather than the full human mind. It retains observable goals, actions, outcomes, conditions, evidence, counterexamples, revisions, and renewed practice. Mao Zedong's *On Practice* can inform the design through its account of knowledge arising from and being tested in practice [7]. Philosophy does not prove program correctness, which instead depends on guards, transactions, and experiments. Long-term-memory research also motivates recovery of task context without equating software fields with human memory [1-2].

### B. Bayesian Components Are Not Experience Itself

Temporal-context models discuss the relationship between retrieval and changing contexts [8]. We use this to motivate environment-aware recovery, not to establish that SQLite structures are human memory. The Thompson-sampling tutorial describes posterior-sampling exploration [9]; our two-arm stationary control covers only its simplest Bernoulli case and cannot establish performance under nonstationarity, partial observability, or delayed real-host feedback.

A Beta posterior describes how evidence changes a belief summary under a working model. It does not establish understanding and does not replace counterexamples, versions, or provenance. It is one numerical component within a lifecycle. Thompson sampling allocates opportunities among verification actions; it is neither the complete formation process nor full reinforcement learning. Allocator performance and lifecycle effectiveness require separate evaluation.

### C. Host AI and Model Training

Runtime experience changes retrievable context, evidence, and policy selection, not parameters. Host AI can inspect candidates, act, and submit feedback, but its free-text responses require separate records of first answers, visible context, tool outcomes, and feedback. The current 15 same-agent, nonblinded exploratory records are not independent model experiments; the 10080 main executions also have no language-model calls. Experience may supply candidate post-training data, but parameter-change claims require fixed pre/post evaluation, forgetting assessment, and rollback evidence. Runtime writeback is not described as self-evolution.

### D. Threats to Validity

Contractual synthetic tasks and deterministic solving reduce model noise but limit external validity. Ten seeds are clusters; 1440 repeated tasks are not 1440 independent samples. Drift and outage schedules are protocol choices, not real-world distributions. Local latency does not equal network or token cost, and same-agent pilots expose previous answers. Interface regression checks verify implementation behavior and are excluded from task-benefit statistics. The end-to-end gate experiment places formation and benefit in one ledger, but its bad claims are a fixed shock imposed by the experimenter rather than an estimate of production frequency; G1 and G2 both reach 100% accuracy on 20 seeds, so the apparatus is insensitive to accuracy and only resolves the two-round outage coverage, the read cost, and first-attempt errors after injection. The real-host experiment faces the non-blinding and the ceiling more directly: the host is also the designer, all 36 answers are correct, and the experiment therefore yields behavioural readings (version admission executed by the host, reuse driving stable-entity reads to zero) rather than an accuracy difference; its arithmetic was executed by a local script, so it cannot be read as evidence about model reasoning either.

## VIII. Conclusion and Future Work

This paper models experience formation as goal-directed action producing observations, comparison with expectations yielding feedback, provenance and condition checking producing candidates, and repeated practice and counterexamples determining stable or disputed states. Versions and restart recovery preserve continued verification. Butian's database, evidence, version chains, context recovery, and feedback interfaces implement observable stages of this process.

The experiments support five bounded conclusions. First, version-admitted rule reuse reduces source reads by 81.60% against exact indexing, while removal of version admission produces 133 obsolete-rule errors. Second, persistence preserves reuse across restart. Third, Beta updating and Thompson controls follow their intended behavior in synthetic numerical environments. Fourth, with the candidate gate returned to the loop, the end-to-end gate experiment reproduces B2-full exactly (1440 correct answers and 253 reads) on the ten original seeds and quantifies the net effect of the gate on 20

seeds: 121 more tasks solved and 2023 fewer source reads (73.33%) than exact indexing, at the cost of 12 additional reads per seed; the state gate blocks bad claims carrying a single evidence item (0 errors versus 209 errors) but not bad claims carrying two independent practices. In the additional dream replay, verified reuse and no dreaming both solve 320 tasks, while verification-inclusive acquisition falls by 37.5%; unverified reuse yields 120 overgeneralization errors. This supports verification before reuse in the two studied families, not unrestricted free-text induction benefits. Fifth, with the same process attached to a real host, all five pre-registered criteria hold over 36 answers: stable entities reach zero source reads in the last two rounds, drifting entities make zero obsolete-rule errors, stable memories rise to warm while drifting ones stay candidate per version; but all 36 answers being correct means the experiment yields behavioural readings only, not an accuracy difference. Costs include more local operations and latency. No claims are made of parameter self-evolution, full Bayesian RL, or general language-model improvement. Future work should connect this same process to real host AI and record frozen first answers, objective tool outcomes, feedback, reuse, and counterexamples round by round to test improvement on related tasks.

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## Appendix A. Implementation Mapping and Reproducible Experiments

The implementation mapping is as follows. Experience storage keeps the current version, parent version, and source event. Practice recording handles success, failure, and dispute transitions. Context composition and parsing retrieve candidate experience and enforce entity, version, and completeness admission. Backup and recovery preserve service state and evidence paths. The mathematical controls implement Beta updating, information gain, utility, and Thompson sampling. The paper explains the key modules, algorithms, constraints, and experimental definitions directly rather than replacing them with an undisclosed static ledger.

The reported experiments consist of deterministic mechanism replay, lifecycle controls, numerical controls, and a dream-effect comparison. Neither the main experiment nor the control experiments call a language model. The dream-effect comparison uses a host-supplied candidate template and produces evidence through real verification and recording interfaces; it is therefore not autonomous discovery or an independent model experiment. These results are mechanism evidence and cannot replace the unfinished confirmatory experiment with a real host AI.

## Appendix B. Additional Experimental Figures

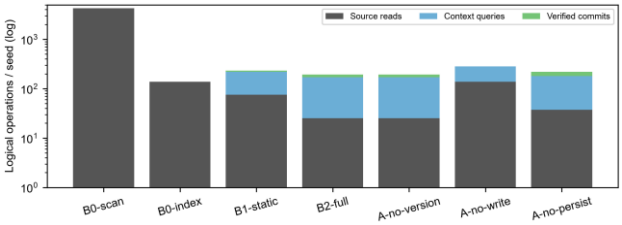



Figure 7. Evidence acquisition, retrieval, and writeback operations in the practice process.

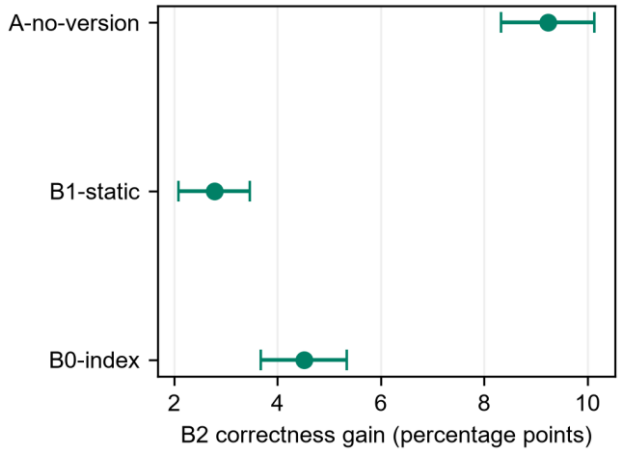



Figure 8. Seed-cluster paired effects.